

A factorial experiment measures the impact of in-store displays on sales for different product characteristics. Variables related to growth or competitive structure are found significant, while market share of the test item in the product category, level of price cut, and advertising to sales ratio have no effect on the impact of display.

Increase in Sales Due to In-Store Display

Supermarkets get a sizable percentage of their total sales through display. Displays are generally set up in piles at the end of an aisle and are often price reduced. Consumers tend to view them as special bargains and often buy from a display a product which they had no previous intention of buying. For the retailer, displays create in-store excitement and increase the average amount purchased. They are generally advertised as "specials" in local advertisements and this method represents a competitive tool supermarket chains use against each other. For example, by displaying high-appeal products at a very low price, it is considered possible to create a low price image for the store as display prices are assumed to be often remembered by the consumers who shop in several supermarkets. The selection, planning, and set-up of in-store displays are an important activity for merchandising managers. Any clues to a better knowledge of the effectiveness of different products on display is certainly of interest to chain managers.

REVIEW OF THE LITERATURE

The sales impact of shelf space allocations has been well documented in recent years [5, 7]. The impact of in-store display of dry grocery products on sales has received less attention. Research by Cunningham and O'Connor [6] using latin square designs led to mixed results. A very comprehensive study by Frank and Massy [11] on in-store promotion did not yield any conclusions because, as the authors mention, "the infrequency of occurrence of displays did not enable

(them) to run any specific statistical analysis" [10]. A recent study by Curhan [8] has thoroughly analyzed the effect of display for fresh fruits and vegetables. It was found that the impact of merchandising and temporary promotional activities is dependent on the product category.

To assess the impact of in-store display for dry grocery products, one has to turn to commercial research. The Dillon Study [9], conducted by *Progressive Grocer*, attempted to measure the increase in sales due to in-store display for the week when the display was set up. An analysis of sales of 360 items on 734 display exposures in 5 supermarkets brought the following results: "Unit sales of the average item featured on display at a reduced price jumped 808%. Dollar sales increased 442%." "Unit sales of the average item featured on special display at a regular shelf price jumped 473%. Dollar sales increased 426%."

The A and P study, conducted by *Progressive Grocer* [1], also indicates for a small number of items, which are not specified by brands, the impact of display on sales. The study found that total advertising and display create "629% more sales than normal" and that display alone (without advertising) creates "a 420% increase in normal movement."

Both studies provide an assessment of the average impact of display and the results may be useful for some specific product categories. But the results do not indicate if the displays were price reduced [1] or whether the display for a given product was with a price reduction or not [9]. The research reports also provide little information on the methodology used and on the statistical validity of the results. Actually, from a methodological point of view, the main problem with those studies is in the selection of the displayed products. If they monitored sales increases due to displays planned in the normal running

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of the stores, then one cannot assess whether sales leaped because the displayed products were carefully chosen by supermarket management for their strong impact or whether the products were selected purely at random. In other words, what they do not deal with is the causality of such leaps in unit sales. Therefore the figures resulting from the survey into display effectiveness do not really give a fair average for all dry grocery items carried by a typical supermarket.

METHODOLOGY OF THE STUDY

The study reported here attempts to measure the display impact, i.e., the increase in unit sales due to display, as compared with regular shelf unit sales for different products. It appeared, through the literature review, that the impact on sales of a display was dependent on the product selected. The tendency for store buyers and merchandising managers to select some products rather than others for display seemed to indicate their bias toward a small number of products and brands. Although promotional allowances are given with some specific frequency pattern to chain stores for most dry grocery products, one always finds displays for the same products and the same brands in competing stores. The first objective of a display study was therefore to select product characteristics that could explain differences in impact of display for different items.

It was necessary to select from a full range of product criteria those that could most likely explain differences in display effectiveness. On the basis of 14 product criteria listed in Table 1, it was necessary to select those that would be considered in the study.

1. It seemed that the position of the product category in its life cycle could most likely explain differences in display effectiveness. It was hypothesized that the visual impact of a display had a stronger effect for new product categories than for mature ones that were already known by most shoppers. The researcher discussed this theory with marketing managers who confirmed that new product categories were the best potential choice for display.
2. It was also hypothesized that the advertising to sales ratio in the product category could help predict display effectiveness. It appeared that product categories that were pre-sold responded in a much stronger way, when displayed, than product categories with a low advertising to sales ratio. This seemed to have been confirmed by the retailer's practice of picking up products in a high advertising to sales category for display.
3. The final product characteristic to be taken into account was the competitive structure and the competitive position of different brands in the product category. It was thought that the presence of several brands with similar market share would lead a category to higher display sales than product categories with a clear brand share leader and small brand share followers. Another theory was that, in the presence of a

Table 1
PRODUCT CHARACTERISTICS CONSIDERED IN THE ANALYSIS

Experiment	<i>Independent variables</i>	
	stage in life cycle	
	advertising to sales ratio	
	competitive structure (leader/follower or close competitors)	
	brand status within category	
	<i>Controlled criteria</i>	
	product movement	→ products were high volume relatively high frequency of use
	price salience	→ products were priced relatively high
	product size	→ products were large packages
	seasonality	→ products were not seasonal items
<i>Variables not tested</i>		
immediacy of use		
impulse purchase		
perceived risk		
number of competing brands		
index of competitive activity		
private label behavior		

strong leader, small brands, in terms of market share, would have little display impact.

The study was, therefore, set up to test the impact of growth rate, advertising to sales ratio, competitive structure within the product category, and brand status within the category. It was decided to test the impact of display for two levels of price cut: a threshold level with a 6% reduction and a deep level with a 12% price reduction. These two levels were said by chain members to be representative of the most regular cuts used for displays. A "two levels of price cuts" design was preferred to a "price cut-no price cut" experiment because it appears that consumers expect products on display to be price reduced [4]. It then becomes more important to analyze how they react to different levels of price cuts rather than to a regular price (which most consumers would view as being slightly underpriced) and a reduced price. It was hypothesized that a deep price cut would lead to a higher sales increase during display than would a threshold price cut.

The different research hypotheses can therefore be summed up as follows:

- H₁: New product categories (more than 5% unit growth) lead to a higher sales increase over regular sales, when displayed, than mature products (less than 5% unit growth).
- H₂: Product categories with a high advertising to sales ratio (more than 5%) lead to a higher sales increase over regular sales, when displayed, than categories with a low advertising to sales ratio (less than 5%).

Table 2
MATRIX USED IN THE FACTORIAL DESIGN

		Competitive structure with leader/follower				Competitive structure with close competitors			
		High A/S		Low A/S		High A/S		Low A/S	
		Growth	Mature	Growth	Mature	Growth	Mature	Growth	Mature
Brand 1	Deep cut								
	Threshold cut								
Brand 2	Deep cut								
	Threshold cut								

H₃: When a product category is made up of close competitive brands, a display leads to a higher increase over regular sales than when a product category is made up of a strong leader and small followers.

H₄: A brand leader increases sales more over regular sales level when displayed than a brand follower.

H₅: A deep price cut (-12%) leads to a larger increase in sales over regular sales for the displayed product than does a threshold price cut (-6%).

These five hypotheses were tested through a factorial design. Other product characteristics were held constant as far as the experiment allowed [3].

THE FACTORIAL DESIGN

The Experiment

The study was conducted in the Boston area in four stores of the First National Stores Company, between January and April 1973. Eight product categories were selected and in each product category, 2 different test brands were studied, with a total of 16 test items. Every week, two displays for different products were set up in each store, making a total of eight displays per week. The four stores were of equal floor space so that the displays in each store would be assumed to have equal impact. A standard location was used for the two displays in each store and a sign "Price Blasted" was set up on the displays.

The structure of the 2⁵ factorial design experiment is described in Table 2. To be able to test each effect with reasonable precision in the analysis of variance, it was necessary to allow an adequate degree of freedom. This called for the setting up of 64 displays (16 test items × 2 price cut levels × 2 replications). In each of the four stores, two products in the same category were always displayed. To avoid carryover effects which would bias the results, four weeks were left between the first display and the second display in the same product category.

Displays were run for each product category as indicated in Table 3. This design enabled product to product comparisons that would minimize store effect.

Displays lasted one week which is the length of time most frequently used by supermarkets. Test product selection was made to meet the characteristics defined in the statement of hypotheses. The factor levels were set at what appeared to be a high or low level for the given criterion, on average and for the whole range of dry grocery products.

Product Selection

Eight product groups were selected to meet the criteria previously mentioned: four mature groups—bleach, mayonnaise, light duty liquid detergent, cooking oil; four growth groups—storage bags, fabric softener, semi-moist dog food, facial tissues. In each group, two brands were selected as test items. In four cases, the products selected had large market share differences; in four cases, they had approximately the same market share. Figure 1 indicates the range of test items selected for study with their associated product characteristics data.

As can be seen in Figure 1, two private labels were used as test items—one in the bleach product group and one in the facial tissues group. Although the use of private labels was not planned prior to discussion with the supermarket chain in which this study was conducted, it does not seem to alter significantly the results. For bleach, no second brand existed in the stores and the private brand appeared a good candidate as a follower. In the facial tissue category, a private brand was preferred to Kleenex because Kleenex had

Table 3
EXPERIMENTAL DESIGN FOR ONE PRODUCT CATEGORY

	Store 1	Store 2	Store 3	Store 4
Week 1	Glad Threshold price	Baggies Threshold price	Glad Deep price	Baggies Deep price
Week 5	Baggies Threshold price	Glad Threshold price	Baggies Deep price	Glad Deep price

Figure 1
PRODUCT SELECTION

Leader-follower	Growth		Regional Market Share		Maturity	Regional Market Share
High A/S	Storage Bags (A/S: 11.3% \$ Growth 37%)	GLAD	46%	Bleach (A/S: 6.2% \$ Growth 2.1%)	CLOROX	68%
		BAGGIES	14%		PRIVATE LABELS	28%
Low A/S	Fabric Softener (A/S: 4.8% Growth: 10.3%)	DOWNY	35%	Mayonnaise (A/S: 2-3% \$ Growth 6.4% unit growth 2%)	HELLMAN'S	38%
		NU SOFT	6%		CAIN'S	21%
Close competitors						
High A/S	Semi-Moist Dog Food (A/S: 7% \$ Growth: 21%)	TOP CHOICE	17%	Light Duty Liquid Detergent (A/S: 7.4%-8.4% \$ Growth: 0.4%)	IVORY	20%
		SPECIAL CUTS	13%		JOY	17%
Low A/S	Facial Tissue (A/S: 4% Growth: 3%-15%)	PRIVATE LABELS	31%	Cooking Oil (A/S: 3.5% \$ Growth: 15% unit growth: 3%)	WESSON	23%
		SCOTT	26%		CRISCO	16%

a very low shelf space and volume in the test stores and would not have been compared as a close competitor to Scott, a strong seller in those stores.

ANALYSIS OF THE RESULTS

On average, for the 64 cases that were part of the experiment, displays with price reductions led to unit sales that were 572% of regular weekly unit sales for weeks preceding the experiment. Between products and treatments, the range varied considerably from 140% to 2345%. The distribution of values appears in Figure 2. In 50% of the cases, sales were less than 400% of normal unit shelf volume and in 25% of the cases, they were less than 300% of normal

unit shelf volume. The mode of the distribution is the interval between 100% and 200%, as can be seen in Figure 2. The distribution is extremely skewed, with relatively few increases greater than 1000% (only 10 cases or 15.5% of the observations).

By product group, the average level of sales, as compared to regular weekly sales, can be seen in Table 4. The analysis of variance of the factorial design was done according to Yates's algorithm [2]. This enables a study of each effect (e.g., price, product category growth) and of the different interactions. Because each observation was replicated once, the computation was carried out on the 32 cell means with the error variance calculated separately from the within-cell variation, as indicated in [12].

Table 5 was computed in this fashion. The dependent variable used was the difference in sales level during display and during regular no-display weeks for 1000 customers. These results were corrected for differences in customer counts because the experiment called for a comparison of results in Week 1 to results

Figure 2
PERCENTAGE INCREASE IN UNIT SALES (TEST VS.
REGULAR SHELF MOVEMENT)
(64 OBSERVATIONS)

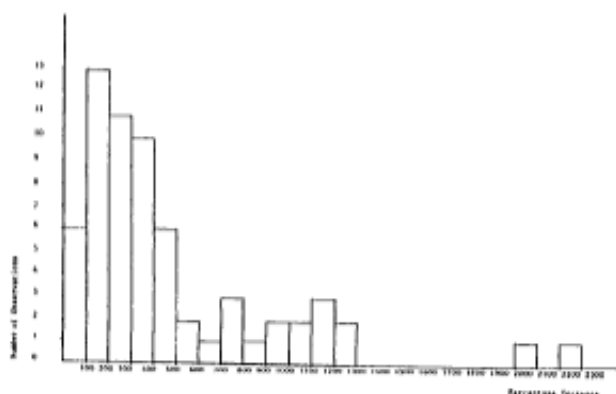


Table 4
AVERAGE SALES DURING TEST WEEK, IN PERCENTAGE
OF REGULAR WEEKLY SALES

Semi-moist dog food	312%
Light duty liquid detergent	1197%
Cooking and salad oil	902%
Facial tissues	321%
Food storage bags	248%
Bleach	331%
Mayonnaise	683%
Fabric softener	580%

Note: Average for 8 observations. No distinction is made here between threshold and deep price cuts.

Table 5
ANALYSIS OF VARIANCE

	Main effect	Sum of squares	F-ratio
<i>Main effects</i>			
A (competitive structure)	-1.11	1.23	5.85 ^a
B (A/S ratio)	-0.24	0.06	0.28
C (growth rate)	-2.32	5.32	25.62 ^a
D (brand)	0.35	0.12	0.57
E (price level)	0.07	0.01	0.05
<i>Two factor interactions^b</i>			
A × B	-1.21	1.46	6.95 ^a
A × C	1.60	2.56	12.19 ^a

^aResults are significant at the .05 level.

^bOther two factor interactions have been omitted because they were not significant. This is also the case of all three factors and four factor interactions.

in Week 5 and it was assumed there were no time variations except those variations taken into account by the customer count ratio.

The analysis of variance indicates at the same time positive and negative results:

1. When they are displayed and price reduced, products belonging to a growth category increase their sales less than products coming from a mature category do. This rejects the first hypothesis. This finding is nevertheless consistent with the product life-cycle approach, which indicates that as a product category matures there is a tendency for marketers to shift their emphasis from an advertising strategy to a more distribution-orientated strategy, e.g., using displays with price reduction. It is generally argued that the reason marketers shift from an advertising approach to a promotional approach is that with the increase in product awareness and with the increase in competition, advertising becomes less and less effective. This study suggests that another reason why marketers use more display when a product category reaches the maturity stage is because displays are then more effective.
2. The competitive structure also has an impact on the effectiveness of display with price reduction. It appears that in competitive structure in which some products have similar positions and in which no one has a clear market share advantage, the average increase in sales, resulting from a display with price reduction, is higher than in competitive structures in which this is not the case. This confirms H₃.
3. The analysis indicates no significant differences among product categories with respect to the advertising to sales ratio. It seems likely that the A/S ratio does not have any direct impact on the effectiveness of display with price reduction. This rejects H₂ and also appears to be relatively contrary to a common rule of thumb in certain supermarket firms. Nevertheless, it is quite difficult to draw general conclusions from the impact of advertising in an in-store display because it is not clear what is the cause or what is the effect. There are two possibilities. Either:

When displays are considered to be effective, the

tendency is for brand managers to use a push strategy and do little advertising. On the other hand, when in-store promotions are considered not effective, the brand managers increase their advertising budgets.

or:

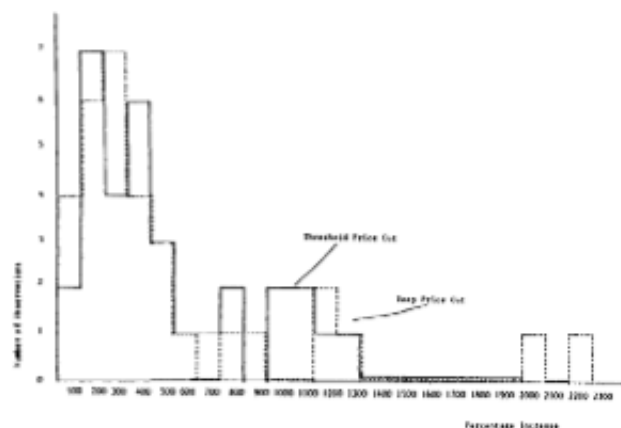
It is possible that, in general, the level of advertising in the product category has no impact on the effectiveness of display.

The second alternative seems more likely because if the first one were true, then we should have a statistically significant reverse relationship. The conclusion seems therefore valid that, in general, the amount of advertising in the product group provides no indication of the impact of display with price reduction.

4. Table 5 indicates no significant differences between increases in sales for deep and threshold price cuts. This is contrary to H₅. This suggests that there is no price elasticity for individual items that are put on display. It seems that consumers expect displays to offer price reduced products and they do not seem to take the time to really compare prices. A tradition of low prices for displays leads people to believe that products on display are generally price reduced. Also, because the dependent variable is sales volume by customer count, it does not reflect possible increases in sales due to more people being attracted into the store.

The results indicate a mean value 553% for unit sales during test week in the 32 cases of threshold price cuts and a mean value of 591% for unit sales during test week in the 32 cases of deep price cuts. On the average, therefore, a price reduction results in a small increase in unit sales 6.9% (not statistically significant) but not in an increase in dollar volume. This is consistent with the Dillon Study. It should be noted that the distribution of the 32 observations for each price level are very different, as can be seen in Figure 3. In particular, in the case of deep price cuts the standard deviation is greater than the mean and the distribution is skewed in the direction of high impact observations.

Figure 3
PERCENTAGE INCREASE IN UNIT SALES (TEST VERSUS REGULAR SHELF MOVEMENT)



5. Results pertaining to the competitive position of a brand in its product category indicate no significant relationship. This was confirmed at the same time in the mean effect and in the two factor interactions ($A \times D$ in Table 5). This rejects H_4 . Within a given product category, it appeared as though the percentage increase in unit sales when a product was displayed would be relatively identical for the two products in the category. This means that a product which sells twice as much as another product when on regular shelf layout should also sell twice as much when on display. In a low-movement product category, this creates a difficult problem for small market share brands; they increase their sales by the same percentage as higher market share brands but their level of sales when displayed is not large enough to qualify their product for a full end display.
6. The only significant two factor interactions are related to the competitive structure in the product category: Although this was not specifically hypothesized, it was found as mentioned in Table 5 that in close competitive structures with low advertising to sales ratio, displays with price reduction are particularly effective. It was also found that in growing markets a leader/follower structure results in a higher display effectiveness (in sales increase) than does a close competitor structure. This effect may be explained by the fact that in a growing market a leader may be able to create awareness for the product group and to encourage new trials (primary demand), whereas close competitors may have a tendency to fight for selective demand gains.

SUMMARY

This research confirms Curhan's findings [7] that there are wide differences among product groups in the immediate increases in unit sales of the displayed products. On the contrary, however, products which belong to the same product category increase their sales, when displayed with a price reduction, by similar percentages. This indicates that in terms of percentage increases in unit sales, products with highly dissimilar market share positions respond similarly to display with price reduction.

It was found that product groups which are growing increase their sales less when displayed than do product groups which are in the mature phase of their life cycle. It was also found that the advertising to sales ratio in the product category had no impact on

the effectiveness of displays. In addition, no differences in display effectiveness could be found for the two levels of price reduction.

This study indicates strong differences between product groups as to their display effectiveness. It has shown the impact of some product category criteria like the growth in the product category or the advertising to sales ratio. It should be considered as a first step in the direction of a better analysis of display effectiveness.

REFERENCES

1. "The A and P Study: Merchandising in Action," *Progressive Grocer*, (1972), 184, 186.
2. Chatfield, Christopher. *Statistics for Technology*. Harmondsworth, Middlesex, England: Penguin Studies in Applied Statistics, 1970.
3. Chevalier, Michel. "The Impact of Display with Price Reduction on Sales Volume and Profitability of Supermarket Products," unpublished doctoral dissertation, Graduate School of Business Administration, Harvard University, 1974.
4. ——— and Ronald Curhan. "Temporary Promotions as a Function of Trade Deals: Descriptive Analysis," working papers, Marketing Science Institute, May 1975.
5. Cox, Keith. "The Effect of Shelf Space upon Sales of Branded Products," *Journal of Marketing Research*, 7 (February 1970), 55-8.
6. Cunningham, Anthony and Norman O'Connor. "Consumer Reaction to Retail Price and Display Changes," *British Journal of Marketing*, 2 (Summer 1968), 148-51.
7. Curhan, Ronald. "The Relationship of Shelf Space to Unit Sales: A Review," working paper, Marketing Science Institute, March 1972.
8. ———. "The Effects of Merchandising and Temporary Promotional Activities on the Sales of Fresh Fruits and Vegetables in Supermarkets," *Journal of Marketing Research*, 11 (August 1974), 286-94.
9. "The Dillon Study," *Progressive Grocer*, (January 1960), 50.
10. Frank, Ronald and William Massy. "The Effect of Retail Promotional Activities on Sales," *Decision Sciences*, 2 (October 1971), 405-32.
11. ———. *An Econometric Approach to a Marketing Decision Model*. Cambridge: M.I.T. Press, 1971.
12. Hicks, Charles R. *Fundamental Concepts in the Design of Experiments*. New York: Holt, Rinehart and Winston, 1966.